

b) $\exists n(n^2=2)$. Let $\exists n(n^2=2)$ be True. $x^2=2 \Rightarrow x=\pm\sqrt{2} \notin \text{set of integers}$
 $\therefore \exists n(n^2=2)$ is False.

c) $\forall n(n^2 \geq n)$. Let $\forall n(n^2 \geq n)$ be False. By defn. of negation, $\neg \forall n(n^2 \geq n)$ is True. By De Morgan's law of quantifiers, $\exists n \neg(n^2 \geq n)$ is True, or $\exists n(n^2 < n)$ is True. Let, for an arbitrary integer, $x^2 < x$.
 $x^2 - x < 0 \Rightarrow \boxed{x(x-1) < 0}$ This can only happen if $x > 0$ and $(x-1) < 0$.
 or, $x > 0$ and $x < 1$. No such integer exist.

This can also happen if $x < 0$ and $x-1 > 0$ or, $x < 0$ and $x > 1$. This is a contradiction. $\forall n(n^2 \geq n)$ is True.

d) $\exists n(n^2 < 0)$. $\exists n(n^2 < 0) \equiv \exists n \neg(n^2 \geq 0) \equiv \neg \forall n(n^2 \geq 0)$ (By De Morgan's laws of quantifiers). By part (a) $\forall n(n^2 \geq 0)$ is True. By defn. of negation, $\neg \forall n(n^2 \geq 0)$ is False. $\therefore \exists n(n^2 < 0)$ is False.

17) Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3 and 4. Write out each of these propositions using disjunctions, conjunctions and negations.

- a) $\exists x P(x) : P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$
- b) $\forall x P(x) : P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$
- c) $\exists x \neg P(x) : \neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$
- d) $\forall x \neg P(x) : \neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$
- e) $\neg \exists x P(x) : \neg (P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$
- f) $\neg \forall x P(x) : \neg (P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

19) Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2, 3, 4 and 5. Express these statements without using quantifiers, instead using only negations, disjunctions and conjunctions.

- a) $\exists x P(x) : P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5)$
- b) $\forall x P(x) : P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$
- c) $\neg \exists x P(x) : \neg (P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5))$